

packages/graph-core/src/embedding-search.ts

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Overview

A new module `packages/graph-core/src/embedding-search.ts` is added.

It introduces a semantic search engine that operates on graph nodes using pre-computed vector embeddings.

Key changes

- **Imports** (lines 1-2) ``ts import type { GraphNode } from "./types.js"; import type { SearchResult } from "./search.js";
- **SemanticSearchOptions` interface** (lines 4-7) ts export interface SemanticSearchOptions { limit?: number; threshold?: number; types?: string[]; } ``-
cosineSimilarity function (lines 14-29) – computes dot product, magnitudes, returns 0 for zero-magnitude vectors.
- **SemanticSearchEngine class** (lines 37-83)
 - Stores `nodes: GraphNode[]` and `embeddings: Map<string, number[]>`.
 - Methods: `hasEmbeddings`, `addEmbedding`, `updateNodes`.
 - `search(queryEmbedding, options?)`
- Applies optional `types` filter, `threshold` (default 0), and `limit` (default 10).
- Computes similarity via `cosineSimilarity`.
- Pushes `{ nodeId, score: 1 - similarity }` for matches.
- Sorts ascending by `score` (lower = higher similarity) and slices to `limit`.

Impact

- New public API: `SemanticSearchEngine` and `cosineSimilarity`.
- Existing code remains unchanged; backward compatibility is preserved.
- Callers must import the new module and ensure that `SearchResult` matches the `{ nodeId, score }` shape returned by `search`.

Risks & follow-ups

- **SearchResult type alignment** – the diff shows `SearchResult` imported but its definition is not visible; mismatch could cause type errors.
- **Default parameters** – `threshold = 0` and `limit = 10` are hard-coded; verify they meet use cases.
- **Zero-magnitude handling** – `cosineSimilarity` returns 0; confirm this behavior is acceptable in all contexts.

- **Sorting logic** – uses `a.score - b.score` ; ensure that lower `score` indeed corresponds to higher similarity as intended.